

Corrigé — Exercice 18

$$1) f(2i) = \frac{2i^2 + 2}{2i - i} = 0. f(1 - 3i) = \frac{5 + i}{1 - 4i} = \frac{1 + 21i}{17} = \frac{1}{17} + \frac{21}{17}i.$$

Idée centrale. $z' - i = \frac{iz + 2}{z - i} - i = \frac{1}{z - i}.$

$$2) E : \operatorname{Re}(z') = \frac{x}{x^2 + (y - 1)^2} = 0 \iff x = 0 \Rightarrow E = \text{axe imaginaire privé de } I.$$

$$F : \operatorname{Im}(z') = 1 - \frac{y - 1}{x^2 + (y - 1)^2} = 0 \Rightarrow x^2 + \left(y - \frac{3}{2}\right)^2 = \frac{1}{4} \Rightarrow F = \text{cercle de centre } \frac{3}{2}i, \text{ rayon } \frac{1}{2}, \text{ privé de } I.$$

$$G : |iz + 2| = |z - i| \Rightarrow y = \frac{3}{2} \Rightarrow G = \text{droite } y = \frac{3}{2}.$$

$$3a) |z' - i| = \left| \frac{1}{z - i} \right| = \frac{1}{|z - i|} \Rightarrow |z' - i| |z - i| = 1.$$

$$3b) \text{ Si } |z - i| = 3 \text{ alors } |z' - i| = \frac{1}{3} \Rightarrow M' \text{ est sur le cercle } \mathcal{C}' \text{ de centre } I(i) \text{ et rayon } \frac{1}{3}.$$